Jeremiah Woods
CE 391 - Lab 3 Report
March 19th, 2026

In this project, Cadence Virtuoso was used to design and simulate an array of 16 6T SRAM cells. Incorporating an iterative design process from schematic to layout simulation, the circuit design was tested through several checks and adjusted as needed. The following tables display the results of the schematic and layout extraction simulations; subsequent pages include snapshots of key design aspects.

Result Tables

Schematic	**Energy Consumption (Single Read and Write, 100MHz)**	**Highest Frequency (Vdd = 1.1V)**	**Point of Failure**
4x4 SRAM Bank	173.4 fJ	500 MHz	Clocked Sense Amplifier

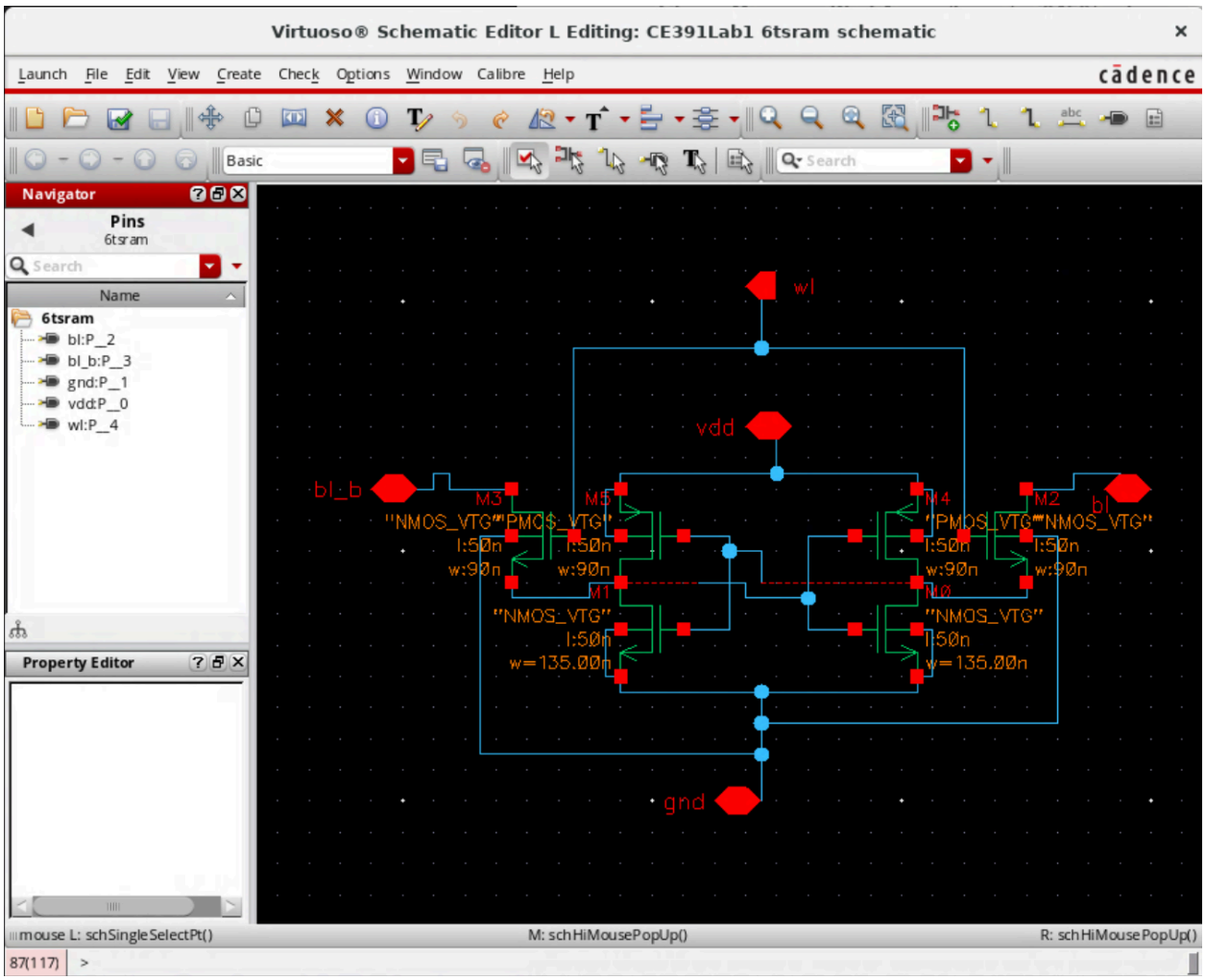
Layout Extraction	**Energy Consumption (Single Read and Write, 100MHz)**	**Highest Frequency (Vdd = 1.1V)**	**Point of Failure**
4x4 SRAM Bank	216.0 fJ	416.67 MHz	Clocked Sense Amplifier

(As requested, the file path is file:///home/qcj9450/cadence/CE391Lab1/6tsram_array and https://moore.wot.ece.northwestern.edu:3300/user/file-manager/cadence/CE391Lab1/6tsram_array.)

Jeremiah Woods
CE 391 - Lab 3 Report
March 19th, 2026

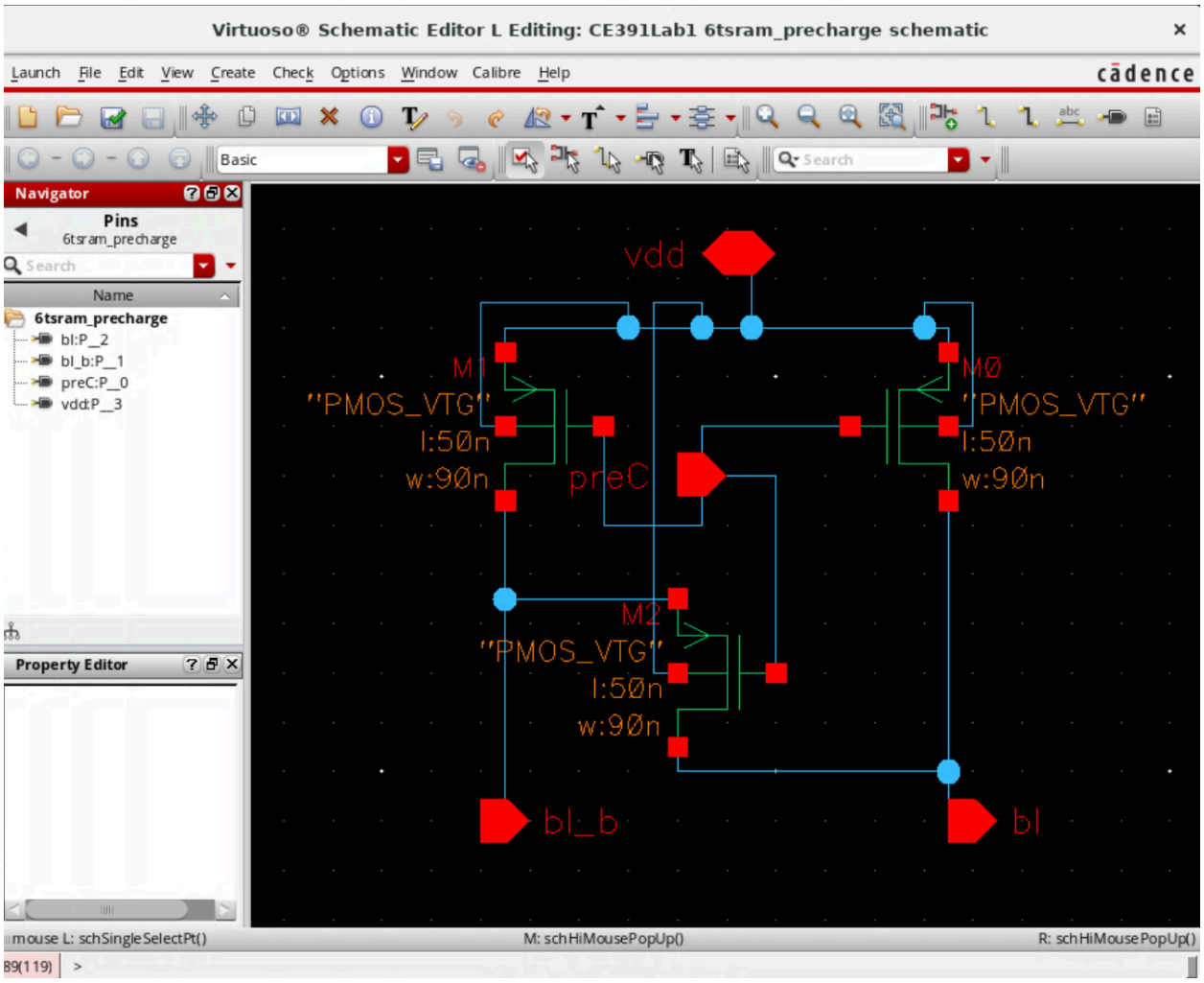
1. Snapshots of Schematics

6T SRAM Cell:



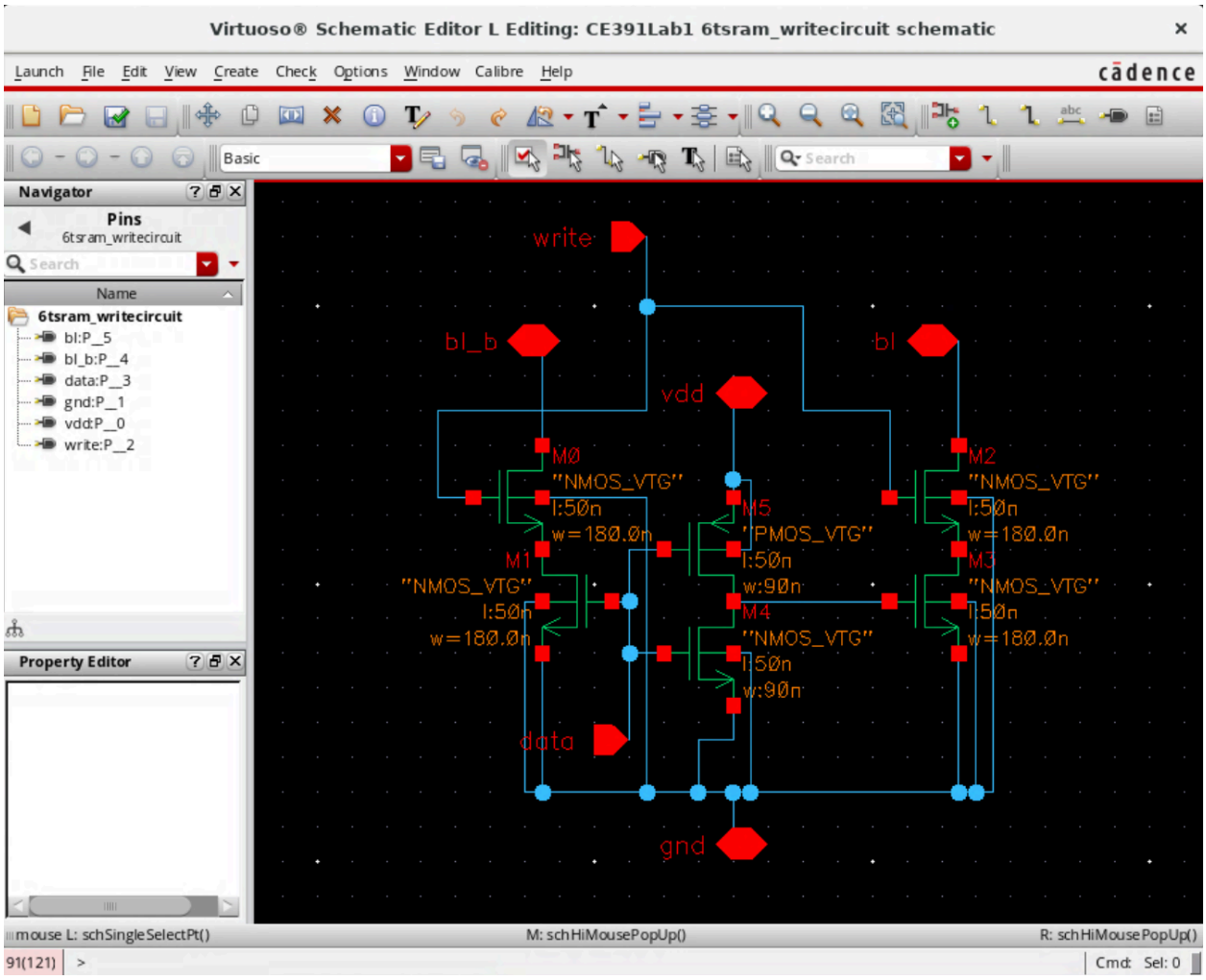
Jeremiah Woods
CE 391 - Lab 3 Report
March 19th, 2026

Precharge Circuit:



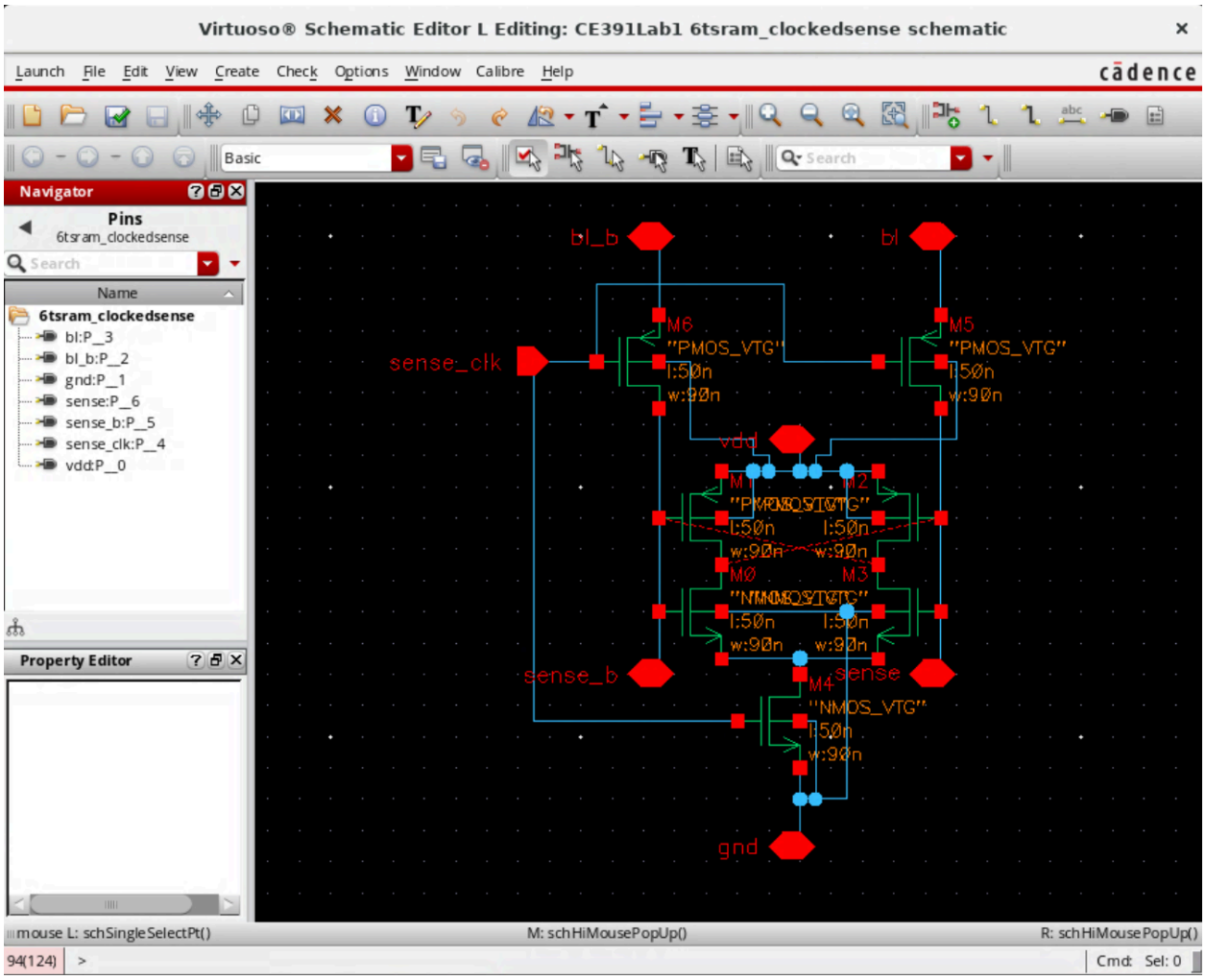
Jeremiah Woods
CE 391 - Lab 3 Report
March 19th, 2026

Write Circuit:



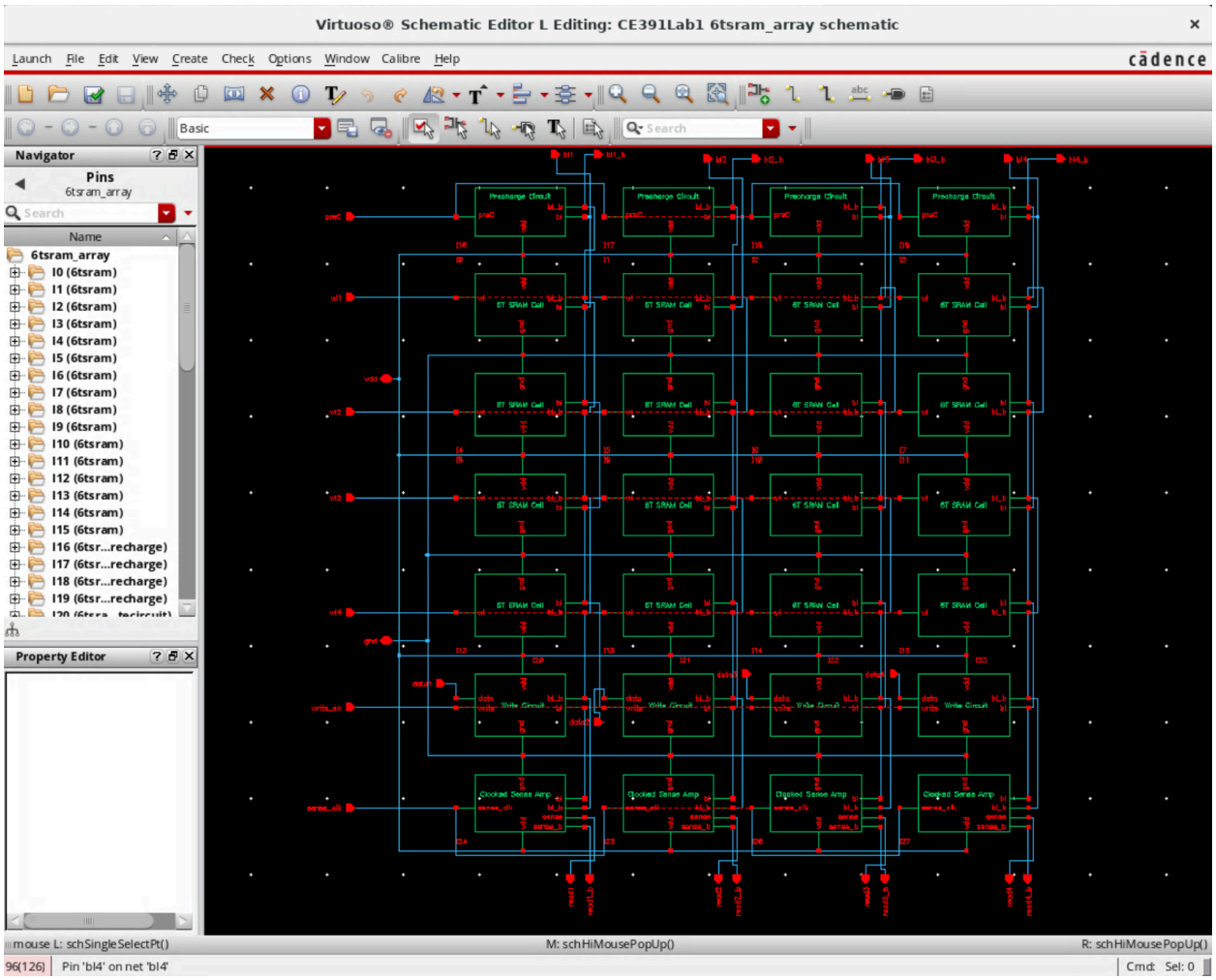
Jeremiah Woods
CE 391 - Lab 3 Report
March 19th, 2026

Clocked Sense Amplifier:



Jeremiah Woods
CE 391 - Lab 3 Report
March 19th, 2026

SRAM Bank:

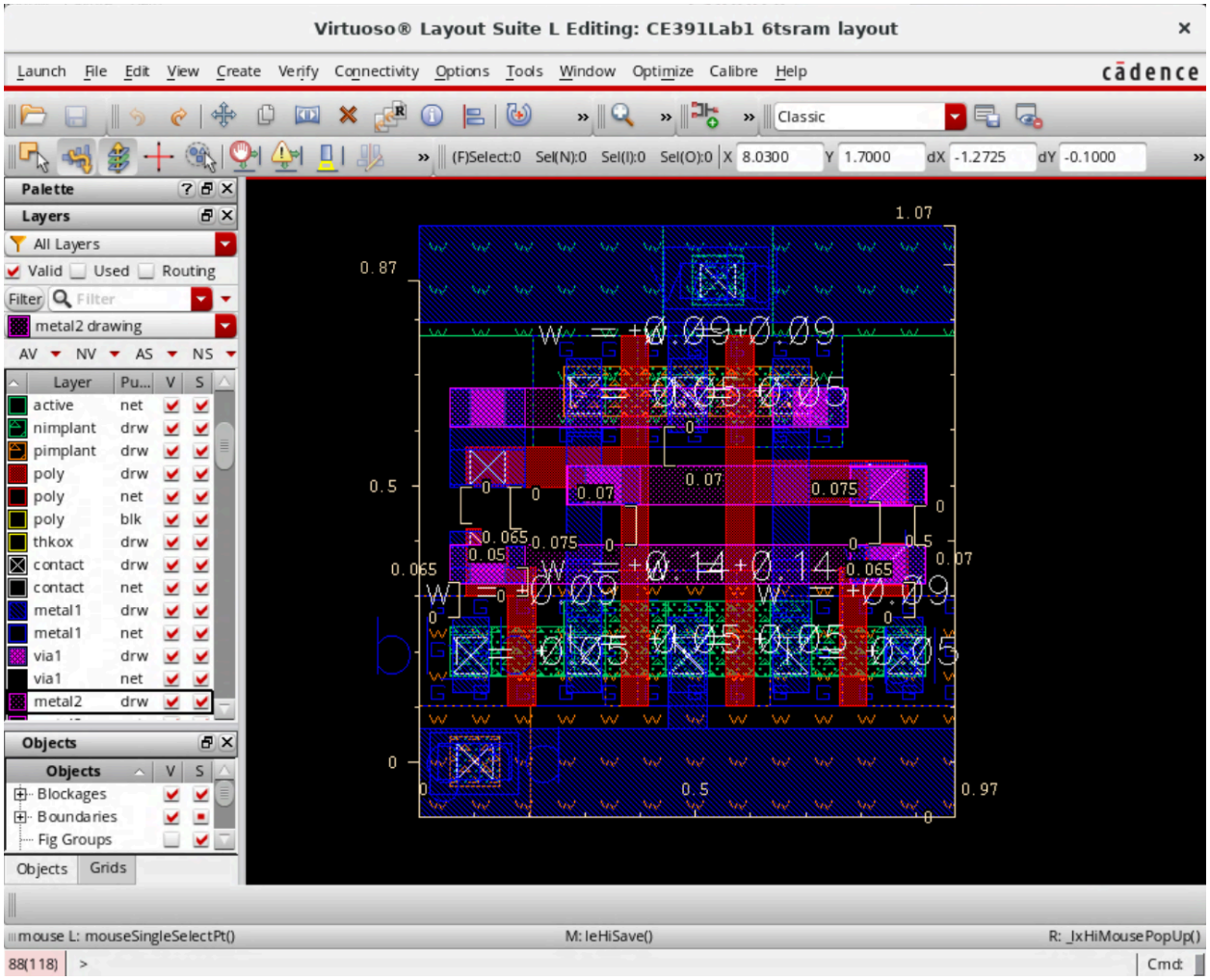


Jeremiah Woods
CE 391 - Lab 3 Report
March 19th, 2026

2. Snapshots of Layouts

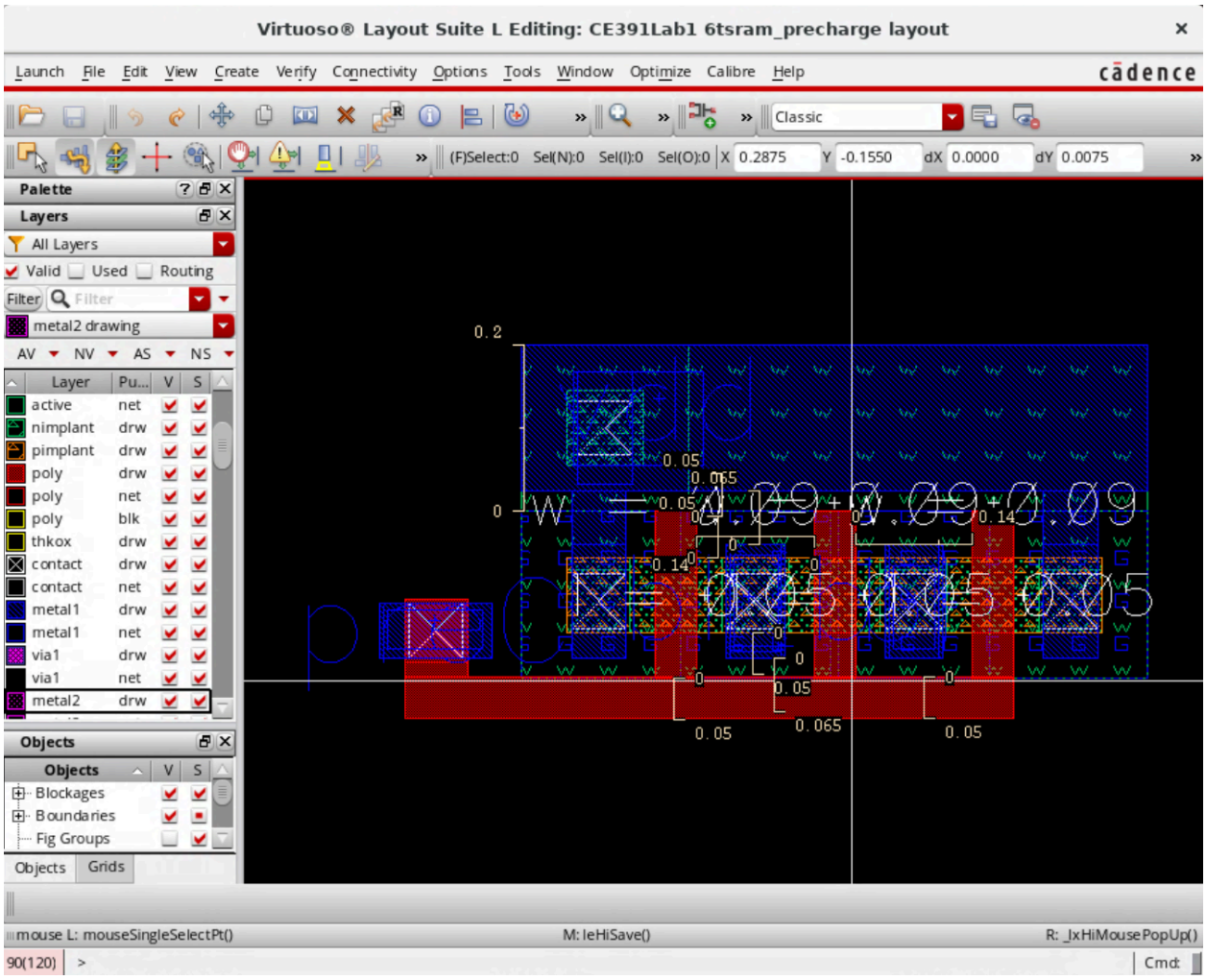
SRAM Cell:

Height = 1.07 um (0.87 from rail center); Width = 0.97 um; Total Area (Center) = 0.844 um^2



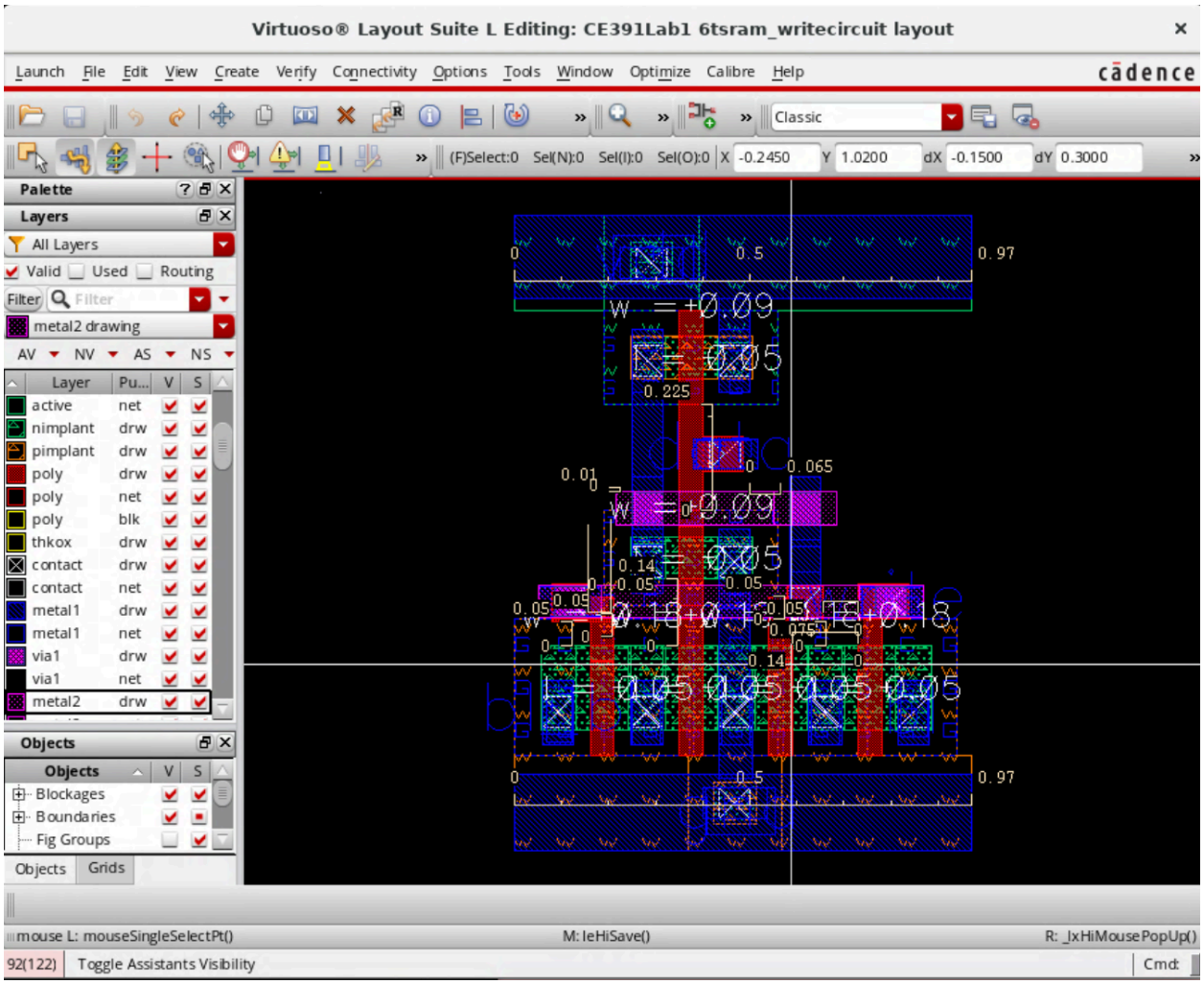
Jeremiah Woods
CE 391 - Lab 3 Report
March 19th, 2026

Precharge Circuit:



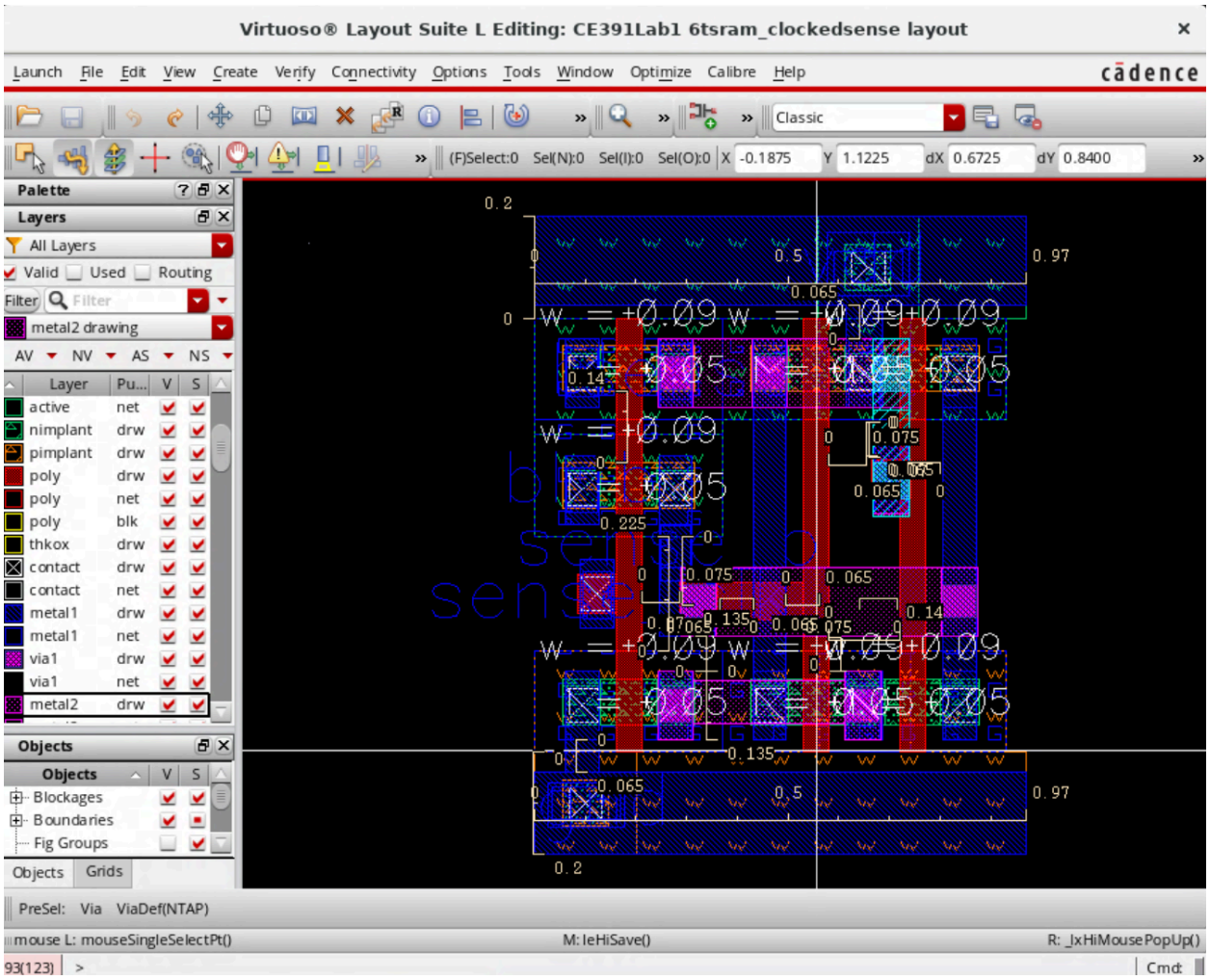
Jeremiah Woods
CE 391 - Lab 3 Report
March 19th, 2026

Write Circuit:



Jeremiah Woods
CE 391 - Lab 3 Report
March 19th, 2026

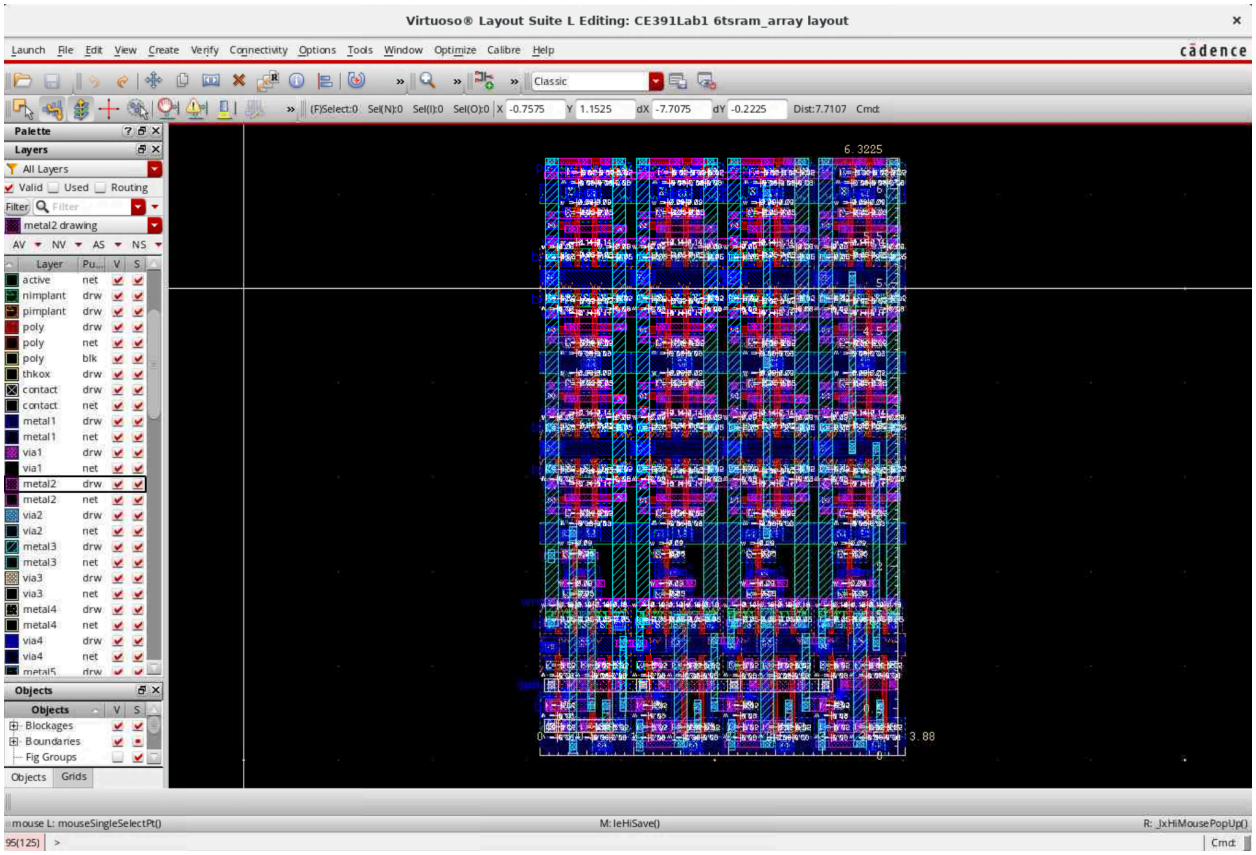
Clocked Sense Amplifier:



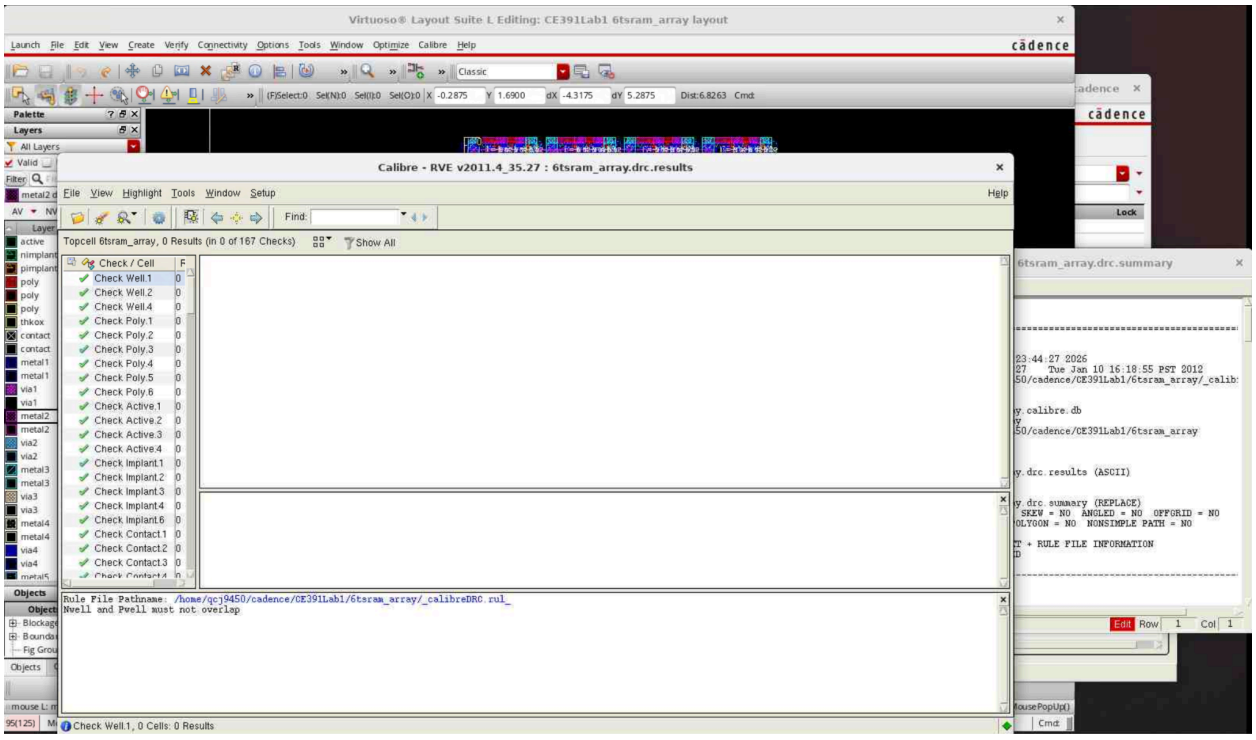
Jeremiah Woods
CE 391 - Lab 3 Report
March 19th, 2026

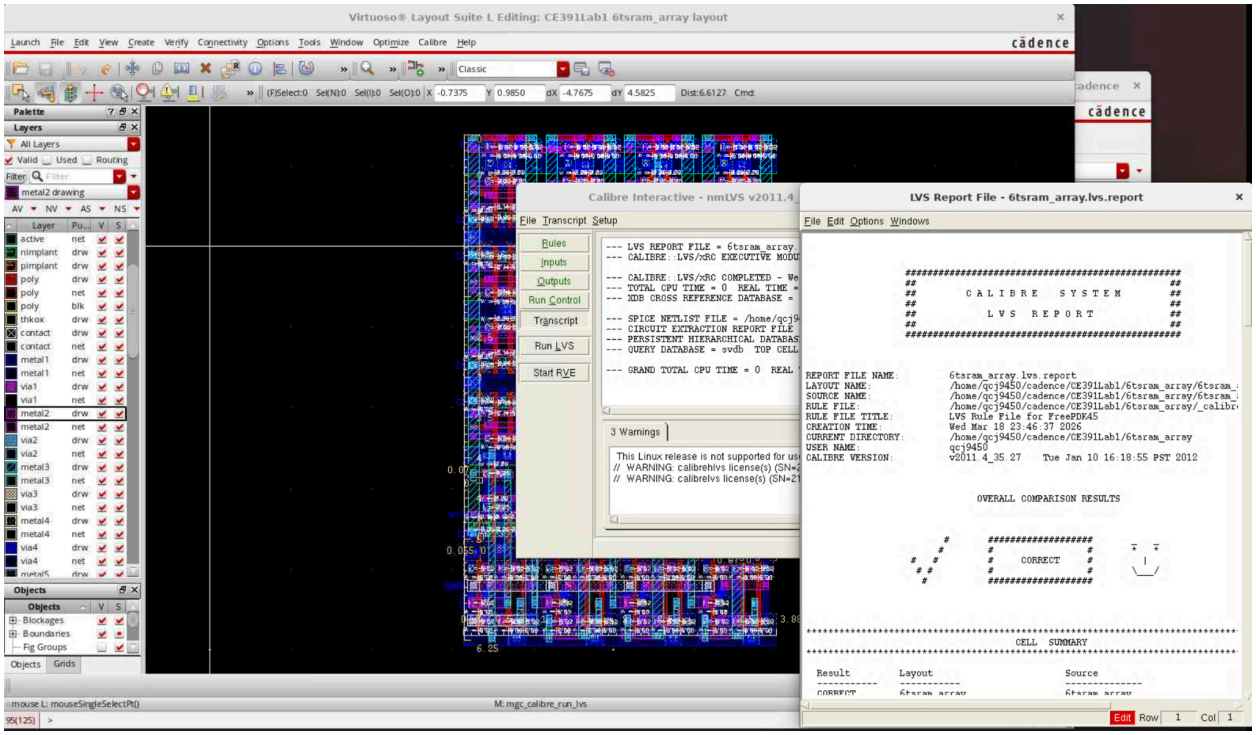
SRAM Bank:

Height = 6.3225 um; Width = 3.88 um; Total Area = 24.531 um^2



3. Snapshots of DRC & LVS Results (SRAM Bank)



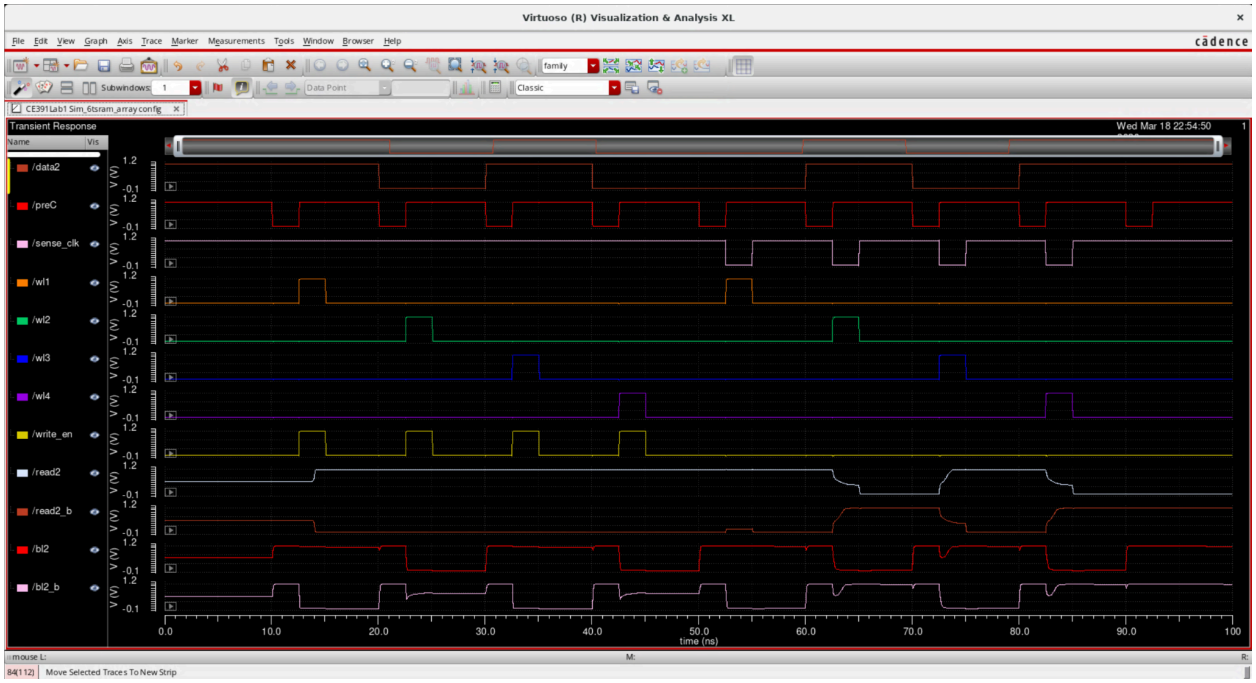


Jeremiah Woods
CE 391 - Lab 3 Report
March 19th, 2026

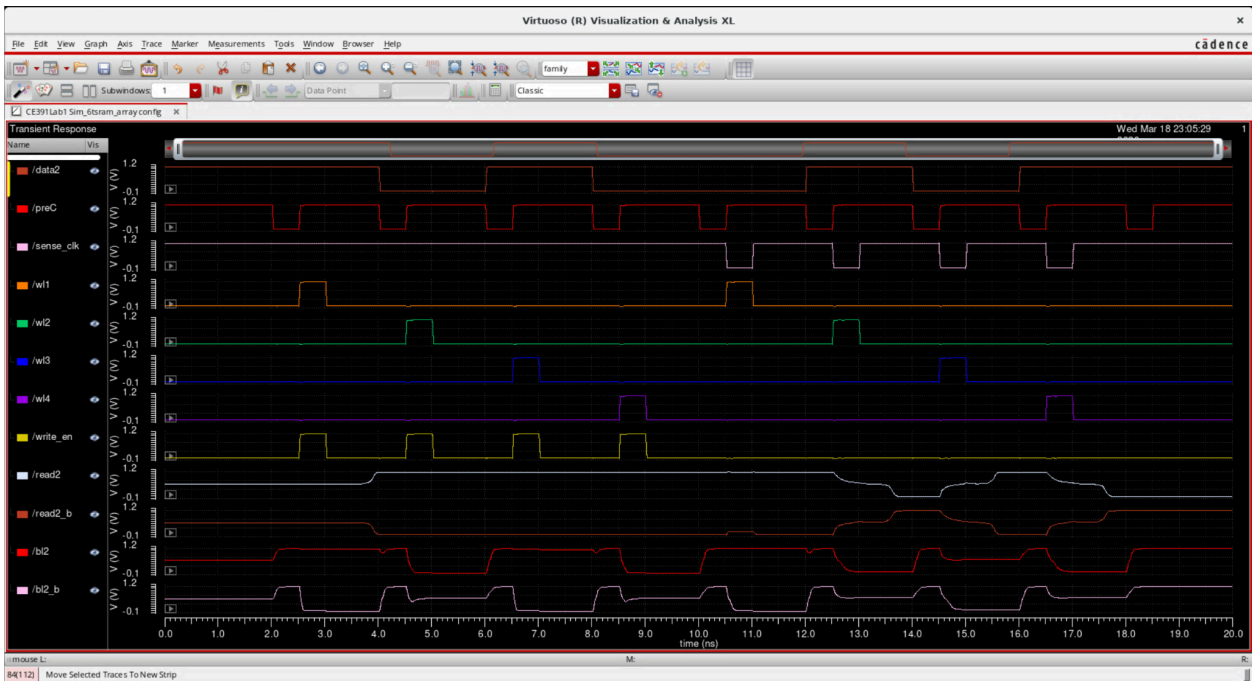
6. Simulation Waveforms

Schematic:

Standard Operation (100 MHz):



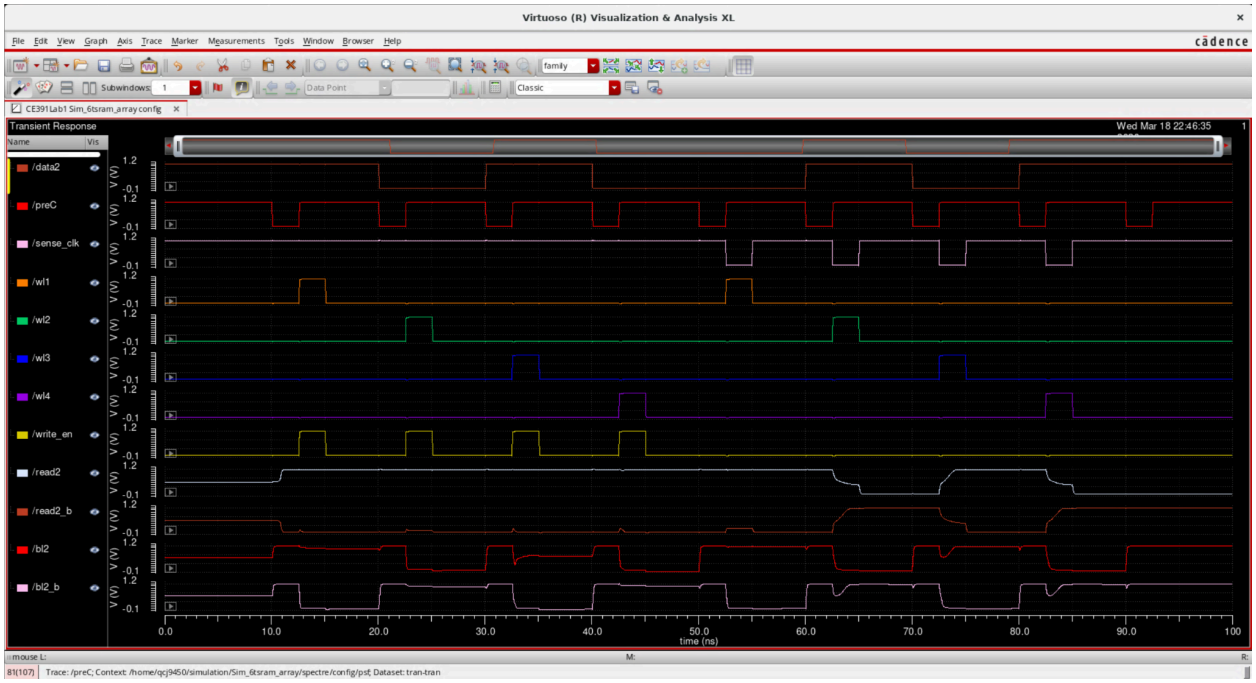
Maximum Frequency (500 MHz):



Jeremiah Woods
CE 391 - Lab 3 Report
March 19th, 2026

Layout Extraction:

Standard Operation (100 MHz):



Maximum Frequency (400 MHz):

